

Exercise 1 SQL fundamentals.

Answers & Output.

1 Select Statement

Syntax

SELECT *

FROM 'workspace'. 'default'. 'employees';

Output.

id	first_name	Last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los A
4	Sarah	Brown	LT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San f
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Doniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-01	San Fran

2 SELECT Distinct (unique Department)

SELECT DISTINCT department

FROM 'workspace'. 'default'. 'employees';

	department
1	IT
2	HR
3	finance
4	Marketing

← Output

3 Order By Statement (Salary by Descending order).

Syntax

SELECT first-name, Last-name, Salary

FROM 'Workspace'. 'default'. 'employee'

ORDER BY Salary DESC;

	first-name	Last-name	Salary
1	Emily	Davis	62 000.
2	Mike	Johnson	60000
3	Robert	Wilson	59 000
4	John	Doe	55 000
5	Sarah	Brown	53 000
6	Daniel	Clark	53 000
7	David	White	52 000
8	Jessica	Moore	51 000
9	Laura	Hall	50 000
10	Jane	Smith	48 000

Output.

4 Limit Statement (TOP 5 Highest Paid Employees).

Syntax

SELECT first-name, Last-name, department, Salary

FROM 'Workspace'. 'default'. 'employees'

ORDER BY Salary Desc

Limit 5;

Output.

	first-name	Last-name	department	Salary
1	Emily	Davis	IT	62 000
2	Mike	Johnson	finance	60 000
3	Robert	Wilson	finance	59 000
4	John	Doe	IT	55 000
5	Sarah	Brown	IT	53 000

5. Where Statement (find All employees in the IT Department.)

Syntax

SELECT first-name, Last-name, department

FROM 'Workspace'. 'default'. 'employees'

WHERE department = 'IT';

	first-name	Last-name	department
1	John	Doe	IT
2	Sarah	Brown	IT
3	Emily	Davis	IT
4	Laura	Hall	IT

Output

6. AND Statement (find employee who work in Finance AND have a greater Salary than 58000.)

Syntax

SELECT first-name, Last-name, department, Salary

FROM 'Workspace'. 'default'. 'employees'

WHERE department = finance

AND Salary > 58000;

Output



	first-name	Last-name	department	Salary
1	Mike	Johnson	finance	60000
2	Robert.	Wilson	finance	59000

7. OR Statement (find Employees that work in the HR or Marketing Department).

Syntax

SELECT id, first-name, last-name, department

FROM 'workspace'. 'default'. 'employees'

WHERE department = 'HR' OR department = 'Marketing';

Output.

	id	first-name	Last-name	department
1	2	Jane	Smith	HR
2	5	David	White	Marketing
3	8	Jessica	Moore	HR
4	9	Daniel	Clark	Marketing.

8. NOT Statement (Employees who do not work in IT Department).

Syntax

SELECT id, first-name, last-name, department

FROM 'workspace'. 'default'. 'employees'

WHERE department != 'IT';

Output

	id	first-name	Last-name	department
1	2	Jane	Smith	HR
2	3	Mike	Johnson	finance
3	5	David	White	Marketing
4	7	Robert	Wilson	finance
5	8	Jessica	Moore	HR
6	9	Daniel	Clark	Marketing

9. IN Statement (People Who Are in HR, IT or finance Department). Syntax

Select id, first_name, lastname, department

from 'workspace'. 'default'. 'employees'

WHERE department IN ('HR', 'IT', 'finance');

Output.

	id	first_name	Last-name	department
1	1	John	Doe	IT
2	2	Jane	Smith	HR
3	3	Mike	Johnson	finance
4	4	Sarah	Brown	IT
5	6	Emily	Davis	IT
6	7	Robert	Wilson	finance
7	8	Jessica	Moore	HR
8	10	Laura	Hall	IT

10. Combining Conditions (find employee who work in the IT department and have salary > 50000

Syntax : SELECT first_name, last_name, department, Salary, City

from 'workspace'. 'default'. 'employees'

WHERE department = 'IT' AND Salary > 50000

AND City = 'New York' ;

Output

First-name	Lastname	department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York.

11. Combining Where AND and Order by Statement

Retrieve the first and last names of employees that work in the finance or Marketing department and earn more than 52 000 and Order by Desc.

Syntax

```
SELECT first_name, Last_name, department, Salary, City  
FROM 'Workspace'. 'default'. 'employees'
```

```
WHERE department = 'finance' OR department = 'Marketing'
```

```
ORDER BY Salary DESC;
```

Output



	first_name	last_name	department	Salary	City.
1	Mike	Johnson	finance	60000	Los Angeles
2	Robert	Wilson	finance	59000	Houston
3	Daniel	Clark	Marketing	53000	Chicago.

12 Combining SELECT DISTINCT, WHERE, and IN

Find all unique cities where employees work excluding those in the IT and HR department.

Syntax:

```
SELECT DISTINCT City, department  
FROM 'Workspace'. 'default'. 'employees'  
WHERE Department NOT IN ('IT', 'HR')
```

Output

	City	department
1	Los Angeles	finance
2	San Francisco	Marketing
3	Houston	finance
4	Chicago	Marketing

13 Combining WHERE, NOT, AND and ORDER BY

Retrieve employees who are not in the finance department having a salary greater than 50000 order by hire date in ascending order.

Syntax.

```
SELECT first-name, last-name, department, salary, hire-date  
FROM 'workspace'. 'default'. 'employees'
```

```
WHERE department != 'finance'
```

```
AND salary > 50000
```

```
ORDER BY hire-date ASC;
```

Output.

	First-name	Last-name	department	Salary	hire-date
1	Emily	Davis.	IT	62000	2015-2-14
2	David	White	Marketing	52000	2016-4-10
3	Jessica	Moore	HR	51000	2018-5-22
4	John	Doe	IT	55000	2021-3-15
5	Sarah	Brown	IT	53000	2021-3-25
6	Daniel	Clark	Marketing	5300	2022-6-1

14 Combining WHERE, OR, IN, and Limit

Find first 3 employees who work in either Chicago or Los Angeles and belong to IT or Marketing department

Syntax

Select first_name, last_name, department, City
from 'Work space'. 'default'. 'employees'

WHERE City IN ('Chicago', 'Los Angeles')

AND department IN ('IT', 'Marketing') Limit 3;

Output

	first_name	Last_name	department	City
1	Emily	Davis	IT	Chicago
2	Daniel	Clark	Marketing	Chicago
3				